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A triboelectric nanogenerator using silica-based powder for appropriate technology

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ABSTRACT

To harvest ambient mechanical energy, a triboelectric nanogenerator is actively researched as a sustainable energy source. One of the advantages of the triboelectric nanogenerator is the almost exclusive use of widely available materials that can be manufactured at a low cost. For example, silica, composed of silicon and oxygen, not only can play a role as a triboelectric material but is abundant everywhere in the earth's crust. Here, we report a triboelectric nanogenerator using three types of silica powder as a freestanding dielectric layer: sand, silicon, and silicon dioxide. Among them, the triboelectric nanogenerator with silicon dioxide powder and polytetrafluoroethylene film produces the highest electrical output power. The instantaneous peak power density is 0.3 mW/m^2, achieved by shaking the triboelectric nanogenerator by hand. Five serially connected commercial light emitting diodes are simultaneously turned on by persons with a hand. The proposed triboelectric nanogenerator can be utilized as a useful electric power generator in Third World due to its low-cost and widely available component materials.

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1. Introduction

Appropriate technology, which helps people in Third World, has been intensively developed. For assuring the basic needs for people in Third World countries, the problems of poverty, unemployment, and inequality need to be urgently solved. Appropriate technology represents simple level technologies for assuring these problems and attaining an efficient result in selected area. It is reasonable for Third World societies to promote less complex and expensive sectors which are helpful to develop the countries. It can increase the level of technologies when the appropriate technology plays a role in a foundation technology in Third World. [1] For example, open source 3D printer designs, which enable low-cost distributed manufacturing system [2], are under development. Additionally, the technology of converting waste plastic into liquid fuel *via* thermal decomposition has been reported to solve waste disposal problems and circumvent the energy crisis [3]. Although those technologies help people in Third World, they cannot supply enough electric energy, which can turn on any electronic device. Therefore, low-cost energy harvesting technology can be favorably utilized for Third World.

Many types of energy are wasted in daily life. However, wasted energy can be transformed into useful electrical energy with the aid of energy harvesting technology. Thus, various types of energy harvesters have been studied using the piezoelectric, [4–6] electromagnetic [7,8], electrostatic [9–11], and thermoelectric effect [12,13]. However, each has their own limitations. A piezoelectric generator suffers from limited material availability, a relatively high-temperature process, and high energy demand in poling process. An electromagnetic generator has its high weight as a weakness, owing to the built-in, heavy magnet. An electrostatic electret generator demands a pre-charging process as an early step during operation. Additionally, a thermoelectric generator requires a large temperature difference to produce electric energy. Moreover, it is difficult to find a practical temperature difference for generating electricity in ambient conditions.

A triboelectric nanogenerator (TENG) was adopted to scavenge ambient energy in recent researches. The TENG device based on a combination of contact electrification and electrostatic induction resulting from the triboelectric effect, which has been known from electrical experiment in 600 BCE The phenomenon of contact electrification is occurred by the simple contact of dielectric-to-metal or dielectric-to-dielectric through friction. Charges move from one material to the other to equalize their electrochemical potential. The electrostatic induction occurs subsequently which is induced from the electric potential difference generated by con-

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tact electrification. The TENG showed several merits, such as wide material availability, simple fabrication with low-cost, light weight, autonomous manufacturing, and broad applicability in ambient environment. [14–21] Moreover, the efficiency of the TENG has been continuously grown [22–26], and eco-friendly technology has attracted considerable attention in recent years [27]. The TENG can be applied as a wearable electric energy generating device and self-powered active sensors [17].

Several operation modes have been developed for each customized applications. [28–30] The powder-based TENG, where the powder is free to move in any direction, was previously demonstrated [31]. The powder plays a role as a freestanding triboelectric dielectric layer, displaying back and forth movements on each other two electrodes [32]. This TENG has several advantages. First, similar to a single electrode mode, it can harvest energy regardless of the direction of vibration [30]. Next, the TENG can avoid the electrostatic shield effect; hence, the charge transfer efficiency can be improved compared to other single electrode-based TENGs comprised of a conventional dielectric film as a triboelectric layer [33].

In this paper, we utilized silica-based particles as a triboelectric material. Silica particles, which are easy to be found in ground, consist of silicon and oxygen. It is known that the elements of silicon and oxygen have weight percentages of 46.6% and 27.7%, respectively, in the configuration of the earth's crust. Particles of Si and SiO_2 with extremely low impurity concentrations were used in two experimental groups, whereas particles of natural sand with relatively high impurity concentration were used in a control group. The representative impurities in the natural sand are Al, Fe, etc. The control group using sand is prepared to evaluate the feasibility of practical use. Among the three different powders, the SiO_2 powder showed the highest output power under the same operation conditions. The electrical signals from the contact of surface between SiO_2 particles and polytetrafluoroethylene (PTFE) film in fabricated TENG show an open-circuit voltage (V_{OC}) of 15 V and a short-circuit current (I_{SC}) of 0.35 μA. By simply shaking by hand, the output power density was approximately 0.3 mW/m^2. Five commercial LEDs serially connected to each other are turned on using the proposed powder TENG, which can be applied for 'appropriate technology'. When a number of this powder TENG with sand particles are used together, sufficient electricity will be supplied to the people in Third World.

2. Experimental

The diameter of the aluminum electrode is 40 mm and the acrylic container has an internal diameter of 32 mm. The height of the electrode and container are 5 mm and 52 mm, respectively. The thickness of the PTFE film is 100 μm. The container is capped by aluminum lids to prevent the powder from leaking. These lids also prevent the external humidity from affecting the performance of the TENG.

The surface characteristics of the Si powder, SiO_2 powder, and sand particles were observed and surface analysis was performed by field-emission scanning electron microscopy (FE-SEM) and energy-dispersive X-ray spectroscopy (EDAX), LEO SUPRA 55 (Carl Zeiss, Germany) with an operating voltage of 10 kV.

An electrodynamic shaker and human hand were used to generate vibration and convert mechanical energy to electrical energy, enabling contact and separation between the dielectric material and counter triboelectric layer with the intensity of 100 N. The output voltage and current generated by the fabricated S-TENG were measured by an electrometer (Keithley6514).

3. Results and discussion

Fig. 1a shows the conceptual illustration of the proposed silica-based powder-TENG (S-TENG). The S-TENG operates as the freestanding triboelectric-layer mode. [17] Three different types of powders, SiO_2, Si, and sand, can serve as a freestanding dielectric layer. Polytetrafluoroethylene (PTFE) surface, which tends to be negatively charged, is located between the powder and electrode. More electrons are transferred to the contact surface of two other materials between SiO_2 and PTFE than between Al electrode and SiO_2. According to triboelectric series, SiO_2 tends to be positively charged than Al. In contrast, PTFE tends to be negatively charged, and larger triboelectric effect at the surfaces between PTFE and SiO_2 occurs than between Al and SiO_2. A thickness of PTFE serving as a counter triboelectric contact surface affects the electrical output of the operating S-TENG device. The value of I_{SC} decreases as the PTFE film attached to the electrode becomes thicker. [34] A thick triboelectric contact surface can cause degradation in electrostatic induction due to the decreased electric field intensity of the contact surface. Therefore, PTFE film layer should be thin. Fig. 1b shows a photograph of the fabricated S-TENG. The diameter of the base side of the container is 40 mm, and its height is 62 mm.

Fig. 1c shows a field-emission scanning electron microscopy (FE-SEM) image of the Si powder and the inset is an image of the Si powder taken by a digital camera. Fig. 1e shows an image of the SiO_2 powder taken by FE-SEM and the digital camera. The particle diameter of the Si powder is approximately 30 μm and that of the SiO_2 powder is in the range of 2 μm–25 μm. The SiO_2 powder represents the non-uniform particle size compared to Si powder. The electrical output can be enhanced with the small particles, increasing contact surface by permeating into the closer position of the contact surface. The weight percent and atomic percent of the Si powder was characterized by energy-dispersive X-ray spectroscopy (EDAX), as shown in Fig. 1d. Fig. 1f represents the EDAX spectrum, weight percent and atomic percent of the SiO_2 powder. Through comparison of the EDAX spectrum of Si and SiO_2, it is confirmed that the elemental composition is obviously different. Moreover, this result illustrates that the SiO_2 powder consists of similar elements with sand particles than Si powder.

The operating principle of the S-TENG is shown in Fig. S1. When the S-TENG is in the initial state, the powder is located at the bottom part of a PTFE film. The powder will be positively charged and the PTFE film will be negatively charged when they come into contact. After shaking, the powder starts to move upward and is separated from the PTFE film. Due to the positively charged silica particles, the induced positive charges are collected at the upper part of the electrode. In this process, the current flows downward. When contact and separation occurs at the upper electrode, it shows the same charge transfer phenomenon which occurred at upper electrode. However, the direction of the current is reversed, because electrons move in the opposite direction.

Fig. 2 shows the open-circuit voltage (V_{OC}) and short-circuit current (I_{SC}) of the S-TENG. The experiment was performed with a vibration frequency at 3 Hz and a 50% volume ratio of the powder in the container. As shown in Fig. 2a, V_{OC} is 5 V and 15 V for the Si and SiO_2 powders, respectively. Fig. 2b shows that I_{SC} is 0.23 μA and 0.35 μA for the Si and SiO_2 powders, respectively. The S-TENG shows 3 times higher V_{OC} and 1.5 times higher I_{SC} using the SiO_2 powder than the output characteristics using the Si powder. Considering of the gap in triboelectric series, silica-to-PTFE shows better triboelectric characteristics than Si-to-PTFE according to the triboelectric series. [35] Moreover, the value of V_{OC} and I_{SC} were enhanced when using the SiO_2 powder, which size of particle is smaller than that of Si powder, as a triboelectric material due to the fact that effective contact area is increased as the size of parti-

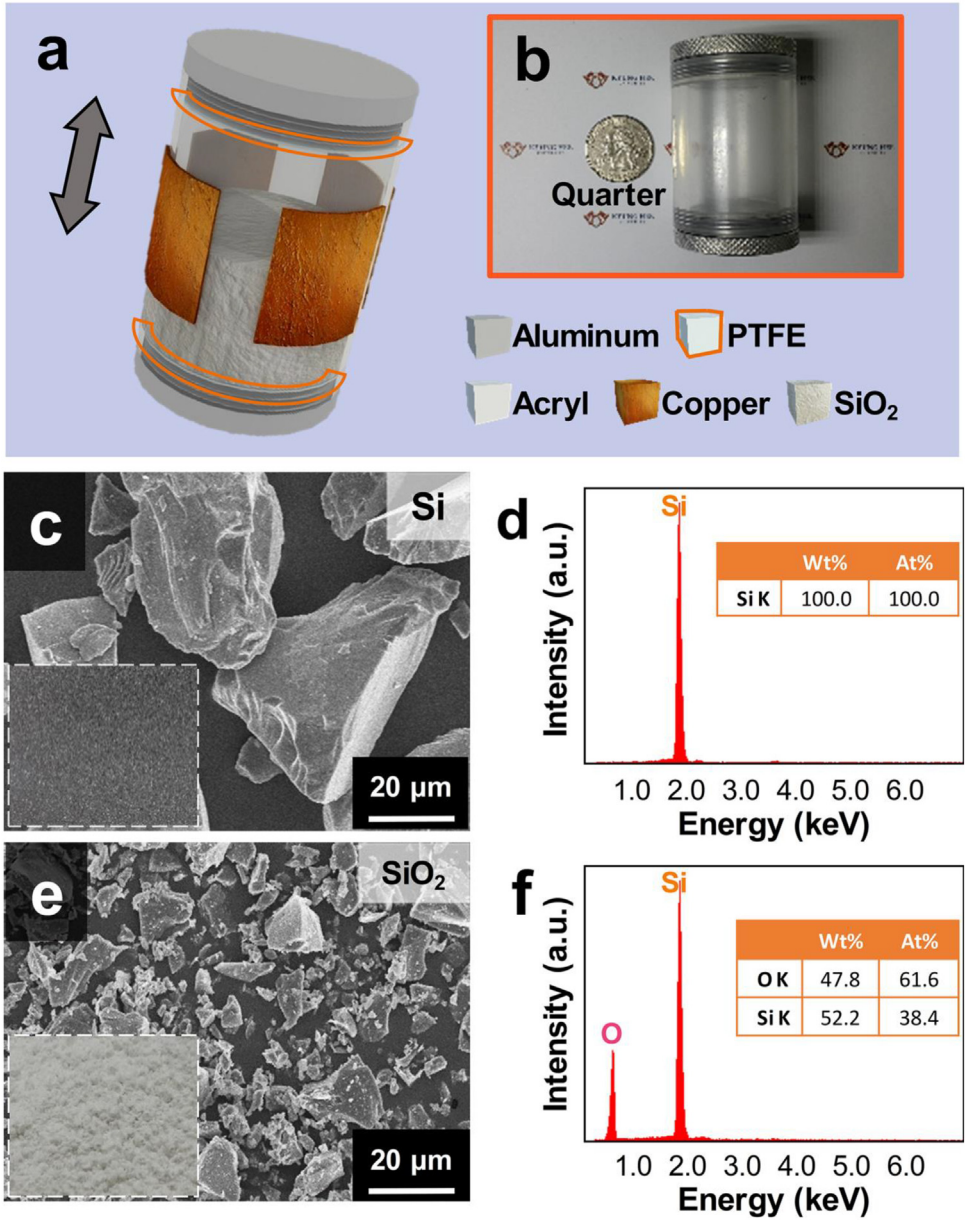


Fig. 1. (a) Cross-sectional diagram of the fabricated S-TENG. This device has an acrylic container and two aluminum electrodes acting as lids. (b) Photograph of the device compared with a quarter. (c) Field emission scanning electron microscope (FE-SEM) image and photograph of the Si powder. (d) Spectrum captured by energy-dispersive X-ray spectroscopy (EDAX) and a table of the weight percent and atomic percent for the Si powder. (e) FE-SEM image and photograph of the SiO_2 powder. (f) Spectrum captured by EDAX and a table of the weight percent and atomic percent for the SiO_2 powder.

cle is decreased. Therefore, the SiO_2 powder was used as a moving material in the container in the following experiments.

To confirm the effect on electrical output using PTFE films, the experiment was conducted under the conditions of 50% volume ratio of SiO_2 powder and shaking at 8 Hz, as shown in Fig. 2c. The circumstance having both upper and lower films shows 1.6 times higher V_{OC} than that with no PTFE films because electrons are more charged when contact occurs between SiO_2 and PTFE than SiO_2 and aluminum. The frequency response of the V_{OC} is shown in Fig. S2. The V_{OC} represents optimum voltage at the low frequency of 5 Hz. When the vibration frequency reaches high frequency exceeding 15 Hz, the V_{OC} is decreased to 30% of the V_{OC} at 1 Hz. At high frequency, the particles are not fully contacted with the surface of PTFE film because the particles move to reverse before contact with the end of one side.

Because of the fluid like properties of the injected powder, it is available to measure the V_{OC} and I_{SC} depending on vibration angle. The angle of the shaker was changed from 0° to −90° at the intervals of 15°, and a vibration frequency was maintained at 3 Hz, as shown in Fig. 2d. The value of V_{OC} decreased as the angle of shaking approaches to −90°. Under this condition, the contact force between the dielectric powder and the lower side film became weak, owing to the collected powder at the bottom of the container. The I_{SC}, which has the peak value of 0.3 μA, shows the same tendency of the decreasing V_{OC}. Photographs of the electrodynamic shaker changing the vibration angle are shown in Fig. S3. The results of the I_{SC} are available in Fig. S4.

The influence of the powder volume is illustrated in Fig. 2e. PTFE films on both sides and shaking at 3 Hz were applied. The V_{OC} showed the highest value in case of the powder volume of 50% of the

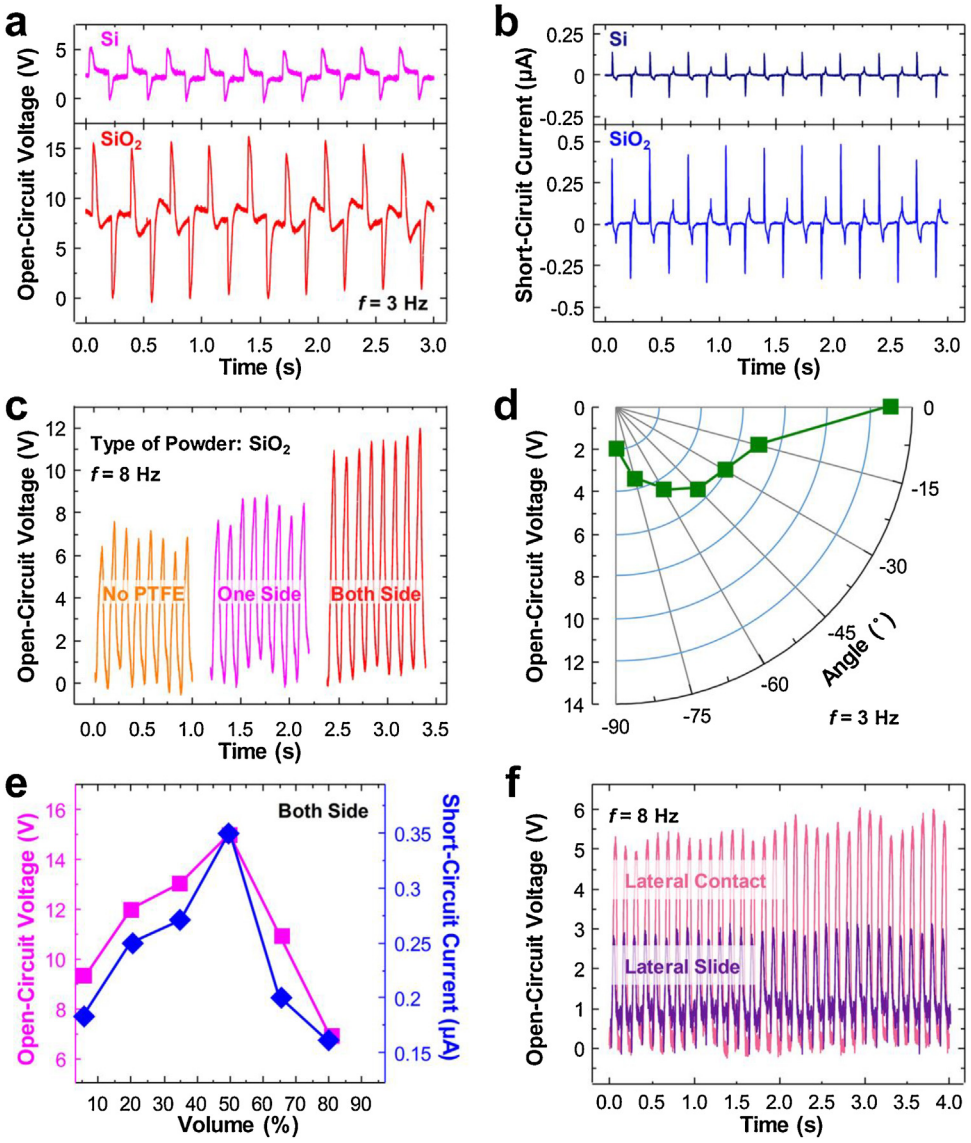


Fig. 2. Electrical output characterization of the S-TENG using the Si and SiO_2 powders. (a) The V_{OC} of the device for the Si and SiO_2 powders. (b) The I_{SC} of the device for the Si and SiO_2 powders. (c) Dependence of the V_{OC} on the number of PTFE films. (d) Dependence of the V_{OC} on the angle of the electrodynamic shaker. (e) Influence of the powder volume ratio on V_{OC}. (f) Output V_{OC} in the case of lateral vibration.

container. This result was caused by the moving distance and contact force of the powder. In case of the powder volume higher than 50%, the distance where the powder can move between electrodes became shorter. This phenomenon induced less positive triboelectric charges on the electrode which is not contacted with particles. [31] Moreover, as the weight of particles increases, more external mechanical energy is required for applying the same vibration before increasing weights of the device. Although the moving distance became longer as the powder volume lower than 50%, the V_{OC} and I_{SC} shows a decreasing tendency due to the smaller contact surface with the less weight and weaker contact force than the case of 50% of powder volume.

For harvesting horizontal vibration energy, additional copper electrodes were introduced. The output voltages depending on the different vibration directions are shown in Fig. 2f. A V_{OC} of 5.3 V in the lateral contact mode and 2.8 V were obtained by each electrode in the lateral slide mode. Electrodes were located perpendicular to the direction of the moving powder in the lateral contact mode. In contrast, in lateral slide mode, electrodes were located parallel to the direction of the moving powder. The contact force at acryl parts, where the copper tape electrode is attached, decreases in comparison to the lateral contact mode, resulting a reduction to 50% in V_{OC}. Photographs of the lateral slide and contact modes can be seen in Fig. S5. Both the lateral contact and slide modes have lower V_{OC} than the vertical vibration mode because the PTFE charges more electrons at the surface of particles than that of acryl. Moreover, the vertical vibration mode also has a longer space where the powder can accelerate speed and force than the lateral slide and contact modes. More accelerated particles increase the contact area and results in increasing of electric potential between the one electrode and another electrode.

We used sand particles as a dielectric material as an alternative to the Si and SiO_2 powders. Fig. 3a shows an FE-SEM image of sand particles with the smallest size among three types of sand. Moreover, the size of the sand particles can be estimated from this FE-SEM image, and the average size is 200 μm. EDAX was also carried out to confirm the composition of the sand particles, as shown in Fig. 3b. This EDAX image shows that the sand particles mainly consist of silicon and oxygen. As a result, the atomic percent of oxygen in sand shows lower than that in SiO_2.

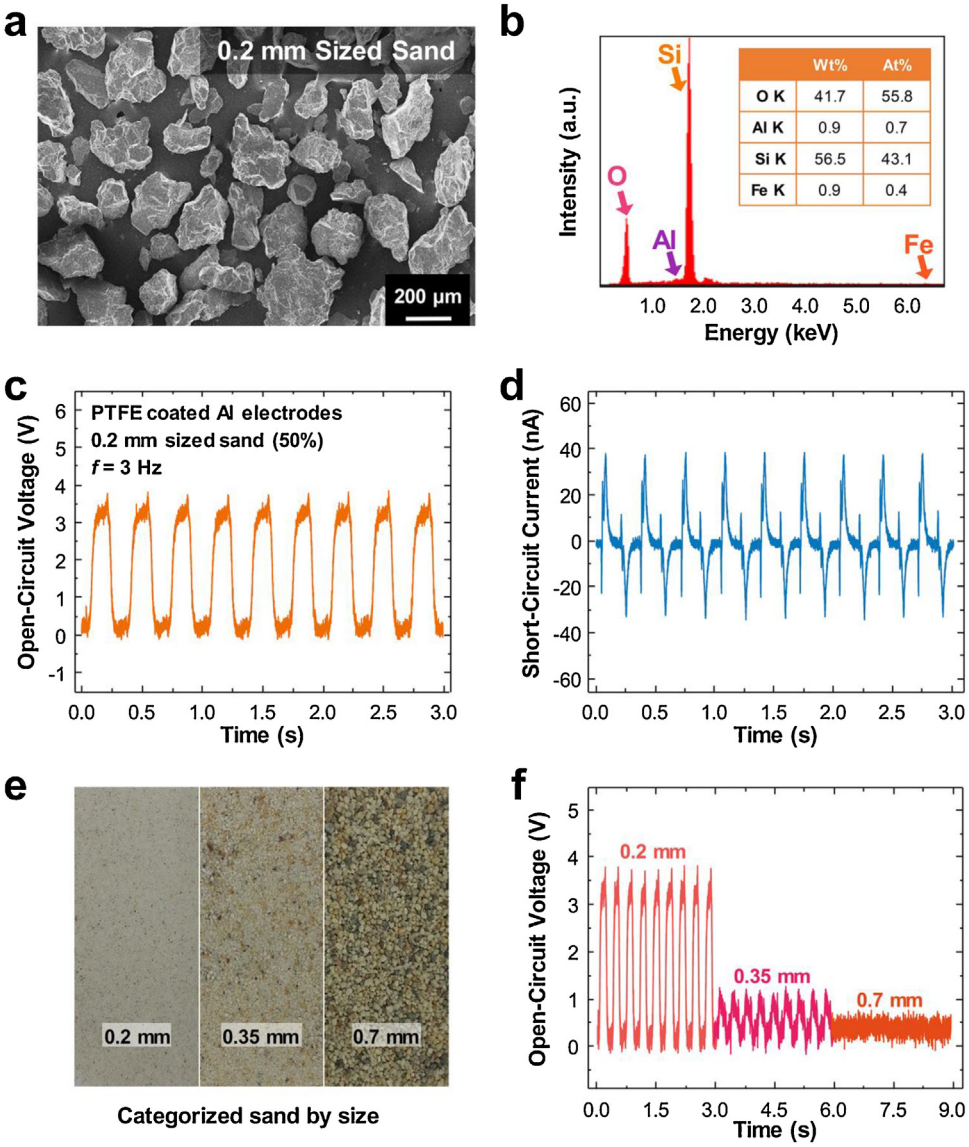


Fig. 3. (a) FE-SEM image of 200 μm-sized sand particles. (b) Spectrum captured by EDAX and a table of the weight percent and atomic percent of the sand particles used in capturing the FE-SEM image. (c) The V_{OC} of the S-TENG using sand particles with a size of 200 μm. (d) The I_{SC} of the S-TENG using sand particles with a size of 200 μm. (e) Photographs of sand particles categorized by the particle size. (f) S-TENG operating with sand particles of different sizes.

In Fig. 3c and d, the V_{OC} and I_{SC} of S-TENG using the group of the smallest sand particles are 3.7 V and 37 nA, respectively. The photographs of sand particles with average sizes of 200 μm, 350 μm, and 700 μm are shown in Fig. 3e. The V_{OC} s of the TENG using the three types of sand particles are 3.7 V, 1.2 V, and 0.5 V, respectively, as shown in Fig. 3f. As the size of the sand particles decreases, the V_{OC} increases due to the fact that smaller sand particles have larger surface contact area during the triboelectrification step. The larger contact area caused their surfaces to induce more triboelectric charges. Consequently, due to the decreased particle size, the greater electrical outputs were generated from the fabricated S-TENG. Interestingly, the S-TENG using sand particles shows the lower electrical output than that using SiO_2 particles. The output voltage will be increased with using the finer sand in aforementioned three types of sands we used in this experiment as a triboelectric material. This result can be derived from the rising tendency of output voltage when the size of particles decreases. A different atomic percent and material inconsistency can also cause degradation in the electrical output. Other components, such as aluminum and iron impurities in sand, could be negatively charged in contacting with the sand and cause the triboelectric charge to decrease in the PTFE film. Moreover, it is confirmed that the mixed sub-micro-sized particles in this sand results in a decrease in the amount of triboelectric charges on the contact surface, which comes from the fact that the sub-micro-sized particles easily adhered to the contact surface due to their light mass and triboelectricity. These sub-micro-sized particles cover the contact surface, resulting in degradation of the output characteristics from the S-TENG. [31]

To further analyze the characteristics of the output power generated from the S-TENG, the output voltage, current density, and output power density of the S-TENG operated by hand shaking, were measured with various external load resistances varying from 0.5 MΩ to 320 MΩ, as shown in Fig. 4a and b. The output voltage is enhanced with an increase of the load resistance. In contrast, the output current density shows the opposite trend against the output voltage, owing to the ohmic loss. The measured current approaches the maximum value as the external resistance decreases to zero. The maximum value of the output power density is the peak value of multiplication of the output voltage and the output current

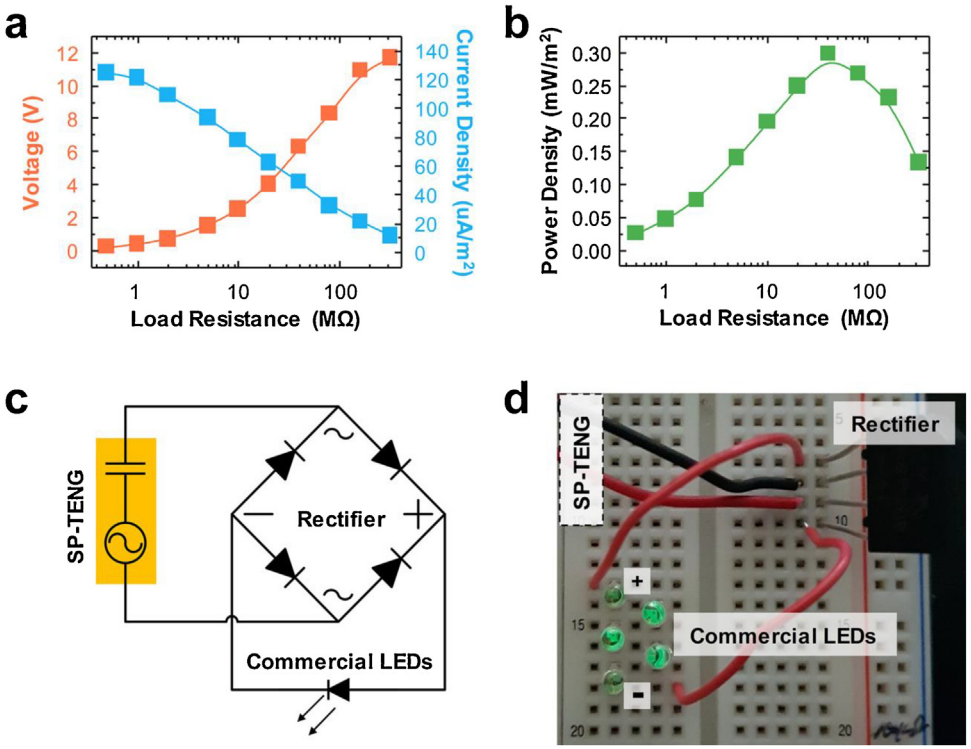


Fig. 4. (a) Load resistance dependency on output voltage and current density. (b) Load resistance dependency in electric power density of the S-TENG in the hand-shaking mode. (c) The circuit diagram for turning on the commercial LED bulbs. (d) Photograph of the illuminated LEDs.

density. The measured maximum power density was 0.3 mW/m^2, corresponding to an external resistance of 40 MΩ.

To demonstrate the performance of the S-TENG in real life, serially connected commercial LEDs were adopted as an electronic device powered by the S-TENG, as shown in Fig. 4c. Commercial LEDs require approximately 2 V for turning on each LED. The S-TENG was shaken by hand in the vertical direction. Five LED bulbs were turned on with flickering as shown in Fig. 4d and Video S1. The stored electrical energy of capacitor connected to the S-TENG is plotted in Fig. S6. The capacitor (0.1 μF) was chosen as a storing device and it requires 15 min to reach 32 V by the fabricated S-TENG with the shaking frequency of 3 Hz. To check the applicability of this proposed S-TENG, commercial drink box with the price of $1 and Al foil was used to fabricate cheaper S-TENG device for the people in Third World. This device is composed of a discarded container with paper body, polyethylene (PE) lid, a pair of Al foils, respectively, as shown as Fig. S7a. Silica particles in discarded container were utilized as a triboelectric material. The external mechanical energy was applied by hand shaking at a frequency of around 4 Hz. The output voltage and current was 9 V and 65 nA, respectively, by this newly fabricated S-TENG. The output plots are shown in Fig. S7b and c. This indicates that the S-TENG can be used for harvesting ambient human motion energy by suspended from the arms and legs. Moreover, the S-TENG can also function as an electric power-supplying device through these energy scavenging characteristics.

4. Conclusion

In summary, the S-TENG can harvest electrical energy regardless of the vibration direction, because this device uses powder, which can move in any direction, as a freestanding dielectric layer. The TENG with a 50% volume ratio of SiO_2 powder produces the highest V_{OC} and I_{SC}. With simple hand-shaking, five serially connected LEDs were turned on. Additionally, abundant electrical energy will be generated when several S-TENG devices are positioned at diverse parts of the human body to collect the ambient energy. Considering the increasing tendency of V_{OC} and I_{SC} in reducing the size of sand particles, the S-TENG using finer sand particles could show higher electrical output than 3.7 V. In Third World, it is possible to fabricate an electric power generator with a discarded container, one pair of aluminum foils, and some sand particles. This S-TENG device can play a role as an effective and inexpensive electric power generator.

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.sna.2018.07.013.

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